taken as a single positive output and the negative terminals are connected simultaneously and taken as a single negative output.

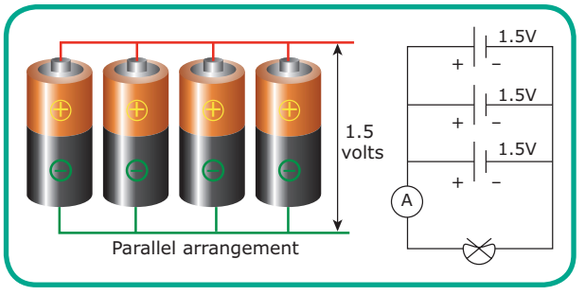
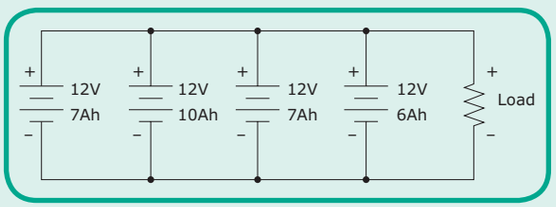


Figure 2.4 Cells Connected in Parallel

Figure 2.4 shows parallel connection of cells. If higher current is needed (having same EMF), cells are connected in parallel. The total EMF is equal to the EMF of individual cells.

Problems

Let four batteries are connected in parallel having a voltage and current rating of 12V/7Ah, 12V/10Ah, 12V/7Ah and 12V/6Ah. Calculate the total current rating of the circuit.



Total current rating of the circuit

$= 7 + 10 + 7 + 6$

$= 30$ Ah

Activity 2.1

- Take few lemons
- Connect the lemons in parallel connection with LED through copper wire
- You can see that LED will glow.

2.2. INDUCTOR OR COIL

Inductor is the passive component used in electronic circuits. It stores energy in the form of magnetic field and delivers it as and when required. An inductor is usually a coil of copper wire wound around a core of ferromagnetic material. Figure 2.5 shows an inductor coil.

DO YOU KNOW?

Who invented the inductor coil?

"Rev.Nicholas Calland" of Ireland invented the inductor coil.

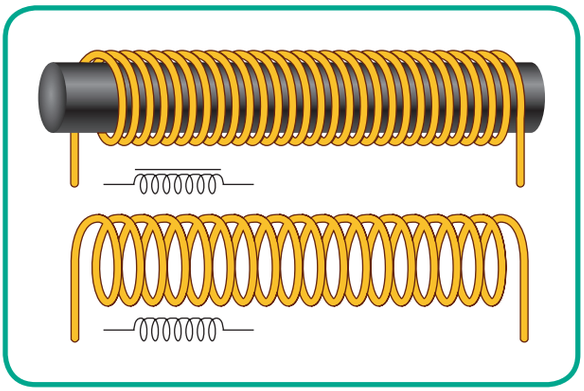


Figure 2.5 Inductors

2.2.1 Inductance

Whenever current passes through a coil, magnetic flux is generated around it. This magnetic flux opposes any change in current due to the induced EMF. The opposition to the change in current is known as inductance. The unit of inductance is Henry.

2.2.2 EMF (Electro Magnetic Force)

Figure 2.6 shows the force acts between the charged particles and the combination of all electrical and magnetic forces. It can be in attractive or repulsive.

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